The State-Aware Enrollment Plane of Post-Quantum ACME: Measurement, Regime Shifts, and Hardened Workflows

ABSTRACT

ACME is a critical control plane for WebPKI and an under-studied locus of post-quantum migration cost. Prior work showed that post-quantum certificate enrollment can be accelerated by alternative re-enrollment paths, but it did not map the state-dependent design space of ACME enrollment or characterize the security cost of such fast paths. To our knowledge, this paper presents the first state-aware enrollment-plane study of post-quantum ACME. We define and evaluate standard issuance (W0), reusable-authorization renewal (W2), certificate-based fast re-enrollment (W3), and a composed migration scenario (S4) across controlled network and load regimes, hostile-edge conditions, and a selective WAN capstone. Across the main campaign, the state-aware lanes substantially outperform W0, with median speedups of roughly 5x for W2 and 4.8x for W3, but their ranking is regime-dependent: W3 is strongest in benign low-latency settings, while W2 overtakes it in hostile regimes even though W3 retains a leaner control plane. We then harden the state-aware lanes with fresh, state-bound proofs, show that the naïve variants admit replay and binding-mismatch scenarios that the hardened variants reject, and complement the executable analysis with a reduced symbolic model. In the hardened slice, most of the performance advantage is preserved, with median overheads around 10% and multi-x gains retained over W0. Our results show that post-quantum ACME enrollment is governed by prior state, workflow geometry, network conditions, and load, not by cryptographic size alone.

CCS CONCEPTS

• **Security and privacy → Network security**; *Web protocol security*; *Authentication*; *Formal security models.*

KEYWORDS

post-quantum security, ACME, WebPKI, protocol measurement, re-enrollment, formal modeling

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1 INTRODUCTION

Post-quantum migration in WebPKI is often presented as a cryptographic replacement problem: larger public keys, larger signatures, larger certificate chains, and new standardized primitives. HTTPS certificates are often studied at the point of use, but their operational life begins earlier, in the enrollment control plane. This plane encompasses request submission, authorization, polling, finalization, certificate retrieval, renewal, and, at times, re-enrollment under operational pressure. Across much of the WebPKI, it is implemented through ACME [1].

This paper treats ACME enrollment as a first-class surface for post-quantum migration: any latency, fragility, or burst sensitivity introduced by post-quantum certificates is incurred before application traffic reaches the TLS handshake. In operational deployments, ACME functions as an automation substrate in which short-lived certificates, fleet-wide renewal, repeated validation, server policy, and bursty request patterns converge within a request-response protocol. The standardization of ACME Renewal Information (ARI) reinforces this operational view, framing issuance and renewal as active systems concerns throughout the certificate lifecycle [4].

The closest prior work is Giron et al., who integrated post-quantum material into Pebble and LEGO and proposed `/pq-order`, a certificate-based fast re-enrollment path authenticated through mutual TLS [5]. Their work established that post-quantum enrollment friction is real and that changing workflow geometry can substantially reduce it. This paper asks the next question: once enrollment is viewed as a stateful system, which state-aware lane should be used, under which operating regime, and under what security contract?

Our thesis is that post-quantum ACME enrollment is governed by four interacting coordinates: prior state, workflow geometry, network regime, and load. Cryptographic size remains important, but it does not order the space by itself. A client with no reusable authorization and no prior certificate is in a different operational state from a client renewing under reusable authorization or re-enrolling through a previously issued certificate. Those states induce different workflows, different request patterns, different polling behavior, and different security assumptions. The right object of study is therefore a *state-aware enrollment plane*: a surface over which workflows change rank as the surrounding regime changes.

This framing leads to three research questions.

RQ1. How much do state-aware ACME enrollment lanes improve post-quantum enrollment relative to fresh bootstrap issuance?

RQ2. Does the fastest state-aware lane remain stable across network and load regimes, or does the ranking change?

RQ3. Can state-aware fast paths be hardened with explicit freshness and binding while preserving most of their performance advantage?

The results cannot be explained by certificate size or request count in isolation. In the broad campaign, both reusable-authorization renewal (W2) and certificate-based fast re-enrollment (W3) substantially outperform fresh issuance (W0), achieving median matched-cell speedups of 5.00x and 4.82x, respectively. Their relative advantage, however, depends on the operating regime. W3 delivers the strongest benign-case performance, including the fastest cell observed in the campaign, whereas W2 overtakes it under hostile-edge conditions despite W3 retaining the leaner control plane. This divergence shows that control-plane minimality is only one component of enrollment performance, and that the fastest design under benign load need not remain fastest once edge conditions deteriorate.

This performance gain also has a security cost. The prior state that accelerates W2 and W3 enlarges their admission surface if it is treated as implicit or enduring authority. Reusable authorization and prior-certificate possession are valuable operational shortcuts, but their authority must remain fresh, scoped, and auditable. We therefore introduce hardened variants, W2H and W3H, based on fresh, single-use, state-bound proofs tied to the ACME account, the normalized identifier set, and the prior-certificate fingerprint. In the selective hardening study, these proofs add roughly 10% median overhead while preserving multi-x speedups over W0. The executable model corroborates the trade-off by showing a sharp reduction in accepted states for the hardened lanes relative to their naïve counterparts.

This paper makes the following contributions.

(1) **A state-aware model of post-quantum ACME enrollment.** We distinguish fresh bootstrap issuance (W0), reusable-authorization renewal (W2), pure certificate-based fast re-enrollment (W3), and a composed migration scenario (S4). This decomposition keeps marginal enrollment mechanisms separate from end-to-end migration journeys, avoiding rankings that conflate a single workflow with a composed operational path.
(2) **A measurement map of the enrollment plane.** We implement a controlled harness over modified Pebble, LEGO, and OQS components, and evaluate a 96-cell wide campaign, a hostile-edge slice, a 12-cell migration study, a 90-cell hardening slice, and a selective WAN capstone. The resulting map shows that state-aware lanes consistently outperform W0, while their internal ordering changes across operating regimes.
(3) **A regime-shift result.** We show that control-plane request count is not sufficient to predict end-to-end enrollment latency. At the hostile edge, W3 sends fewer requests than W2 but incurs higher latency, revealing a regime shift driven by the interaction of workflow geometry, payload shape, synchronization structure, and load.
(4) **A hardened design point for state-aware enrollment.** We introduce `authzReuseGrant` and `pqReenrollmentToken` as fresh, single-use, state-bound proofs for W2H and W3H. Unit tests, an executable state-space model, and a reduced symbolic model support the claim that the hardened lanes reject replay and binding-mismatch scenarios that the naïve variants admit or leave under-specified.

The broader lesson is that post-quantum WebPKI migration reaches beyond primitive replacement. Enrollment is itself part of the transition: state-aware lanes can deliver substantial operational gains, but only when their state assumptions are explicit, measured, and hardened.

2 BACKGROUND AND RELATED WORK

ACME as a WebPKI control plane. ACME automates certificate issuance through a small set of HTTPS resources: directory discovery, account creation, order creation, challenge validation, polling, finalization, and certificate retrieval [1]. Identifier validation is challenge-based, with `http-01` and `dns-01` in RFC 8555 and `tls-alpn-01` standardized separately in RFC 8737 [1, 14]. Renewal is not a distinct base-protocol transaction, structurally, renewal is another order, and any advantage from reusable state depends on server policy and implementation behavior.

This focus is warranted because ACME is the automation layer behind large-scale certificate lifecycle management. Its cost spans cryptographic work together with round trips, polling, authorization-reuse policy, challenge-service exposure, server-side queues, and bursty request arrival patterns. The standardization of ARI as RFC 9773 reinforces this operational view: issuance and renewal remain active engineering problems across the certificate lifecycle [4].

Post-quantum migration in WebPKI.. The post-quantum transition has moved from primitive selection to systems migration. NIST has standardized ML-KEM and ML-DSA [8, 9], but primitive availability does not settle deployment cost. Larger keys, signatures, and certificate chains can affect transport, parsing, caching, renewal, and recovery. Prior work on hybrid X.509 certificates studies certificate representation and compatibility during the transition [2]. Measurement studies of post-quantum and hybrid TLS show that protocol performance depends on both primitive cost and message structure [12, 15]. KEMTLS and related work further show that changing handshake geometry can be as important as changing algorithms [3, 13].

Post-quantum ACME.. Giron et al. are the closest antecedent [5]. They integrated post-quantum and hybrid material into Pebble and LEGO and proposed `/pq-order`, a fast re-enrollment path for clients already holding a certificate. Their key insight was that post-quantum enrollment can be accelerated by changing workflow geometry, not only by optimizing cryptographic operations.

This paper extends that line in three ways. First, it compares multiple state-aware enrollment lanes rather than only standard issuance and one certificate-based fast path. Second, it treats network regime and load as first-class axes, showing that the ranking between state-aware lanes changes under stress. Third, it addresses the security cost of state-aware acceleration by adding fresh, state-bound proofs and measuring their performance overhead.

3 STATE-AWARE ENROLLMENT MODEL

3.1 Enrollment as a Stateful Plane

The key modeling choice is to treat enrollment as a function of prior state and workflow choice. Let

$$F(\sigma, w, \kappa, \eta, \lambda) \to Y$$

denote an enrollment experiment, where σ is the client's prior state, w is the workflow, κ is the cryptographic profile, η is the network regime, λ is the load regime, and Y is the observed outcome vector: latency, requests, bytes, polls, phase timing, and selected system metrics.

We summarize prior state as

$$\sigma = (A, Z, C, M),$$

where A indicates an existing ACME account, Z reusable authorization state, C a prior certificate for the target identifier set, and M admissibility of that certificate for mTLS-based re-enrollment. The purpose of such notation is to prevent workflows with different state assumptions from being compared as though they began from the same condition.

Table 1 defines the measured objects. W0 is the canonical baseline. W2 and W3 are pure state-aware lanes, but they consume different state. S4 is a composed migration scenario. It is retained because operators care about the cost of getting into a state-aware lane, but it is kept outside the pure-workflow ranking.

3.2 Security Assumptions and Threats

State-aware acceleration therefore carries a corresponding security obligation. Once a CA enables W2 or W3, enrollment decisions incorporate prior state: reusable authorization in W2 and prior-certificate possession in W3. This state must be sufficiently fresh, bound to the correct ACME account and normalized identifier set, and consumed only under conditions that the CA still accepts.

Our threat model covers an adversary that can replay previously observed proof material, submit expired proof objects, mismatch account identifiers, request a different identifier set, present or reference a different prior certificate, or invoke a fast lane when the relevant reusable state is inactive. We exclude compromise of the CA signing key. For W3, compromise of the client private key corresponding to the prior certificate is also outside the protection goal of the token itself, since mTLS possession is part of the trust assumption for certificate-based re-enrollment. The hardening goal is that stale, replayed, or weakly bound prior state must not be accepted as current authorization.

Table 2 summarizes the hardened design. In W2H, `authzReuseGrant` serves as a short-lived, single-use proof authorizing consumption of reusable authorization state. In W3H, `pqReenrollmentToken` plays the analogous role for the `/pq-order` transaction. Both proofs are bound to the same core tuple: the ACME account, the normalized identifier set, and the prior-certificate fingerprint. When prior state enables acceleration, its consumption must require fresh, single-use evidence bound to that state.

3.3 Cost Model

For interpretation, we use the following decomposition:

$$T_{\text{total}} \approx N_{\text{flights}} \cdot RTT_{\text{eff}} + \frac{B_{\text{wire}}}{BW_{\text{eff}}} + C_{\text{client}} + C_{\text{server}} + Q + \epsilon. \quad (1)$$

The terms in Equation 1 decompose enrollment cost into workflow geometry, on-wire payload, endpoint computation, and queueing or contention through Q. The equation isolates the mechanisms that can dominate latency under different regimes, without claiming instruction-level or implementation-specific precision. This framing captures that reducing the number of control-plane requests does not guarantee lower end-to-end latency when payload cost, computation, synchronization, or contention becomes the limiting factor.

4 EXPERIMENTAL DESIGN

4.1 Implementation and Harness

The artifact builds on Pebble as a compact ACME CA, LEGO as the client, and Open Quantum Safe libraries for post-quantum primitives [6, 7, 10, 11]. The benchmark harness orchestrates state seeding, workflow execution, network profiles, load regimes, repetitions, measurement harvesting, and export into canonical CSV files. Instrumentation records end-to-end latency, HTTP request counts, authorization polls, HTTP bytes, phase-level timings, structured request traces, and selected system metrics.

The harness enforces the semantic separation in Table 1. For W2, the measured phase begins after reusable authorization has been seeded. For W3, it begins after the prior certificate needed for mTLS is available. S4 explicitly includes a seeding phase followed by transition through the fast path, and is therefore reported as scenario cost.

4.2 Study Matrix

Table 3 summarizes the harvested studies. The main campaign crosses

$$\mathrm{W0/W2/W3} \times \mathrm{C0/C1/C3/C5} \times \mathrm{N0/N2/N3/N4} \times \mathrm{L0/L1},$$

for 96 cells. The hostile-edge study fixes C5/N4/L2 and compares W0, W2, and W3. The migration study evaluates S4 over C1/C3/C5 and N0/N2/N3/N4 at L0. The hardening study compares W0, W2, W2H, W3, and W3H over a targeted slice of crypto, network, and load regimes. The WAN capstone evaluates W0, W2, and W3 under C0 and C1 over a real path between a local client site and a Frankfurt-region CA host.

Table 4 defines the profiles and regimes used throughout the evaluation. The OQS profile names are retained as artifact labels so that each result remains traceable to the measured configuration. Within this experimental scope, the evaluation uses these profiles as a controlled vocabulary for studying workflow and regime effects; exhaustive coverage of every post-quantum or hybrid certificate construction lies outside the claim.

4.3 Metrics and Comparisons

The primary ranking object is matched-cell speedup versus W0. A matched cell holds crypto profile, network regime, and load regime fixed, changing only the workflow. This prevents a state-aware lane in a benign environment from being compared to a baseline in a harsher one.

Medians are the primary summary statistic for an enrollment plane that spans benign local cells, emulated high-latency settings, bursty workloads, and hostile-edge stress. We additionally report p95, minima, and maxima where they expose tail behavior, extrema, or changes in surface shape. Request and byte summaries are treated as cell-level observables under the configured load regime. Consequently, the L1 and L2 request counts describe aggregate concurrent cell execution at cell granularity, comparisons with a single isolated enrollment must account for that aggregation.

4.4 WAN Capstone

The WAN capstone serves as a confirmatory slice of the evaluation. The CA runs on a Frankfurt-region host, while the client remains

Table 1: Semantic contract of the measured workflows and scenario. The paper ranks only the pure workflows. S4 is retained as scenario-cost evidence for migration operations.

Lane	Operational role	State consumed	Measured scope
W0	Standard bootstrap issuance	No reusable authorization and no prior certificate required	Account use, new order, `http-01` challenge resolution, polling, finalization, and certificate retrieval.
W2	Reusable-authorization renewal	Reusable authorization state already seeded	Renewal pass after reusable authorization exists. The seeding phase is outside the measured lane.
W3	Pure `/pq-order` fast re-enrollment	Prior certificate available for mutual TLS	Fast re-enrollment and certificate retrieval after the prior certificate is already provisioned.
S4	Composed migration scenario	Fresh challenge followed by prior-certificate mTLS	Scenario cost: seed a classical certificate, then transition through the fast path. It is not ranked as a pure workflow.

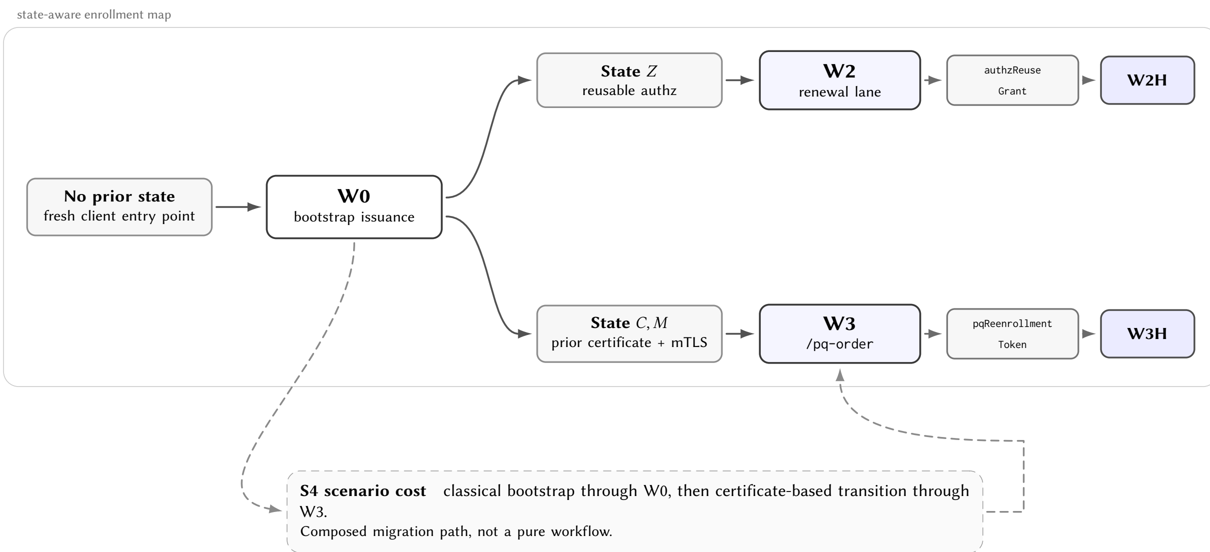

Figure 1: Workflow-state map. The figure separates bootstrap issuance, reusable-authorization renewal, certificate-based fast re-enrollment, and composed migration cost. Hardened gates attach fresh proofs to the state-aware lanes.

Table 2: Binding requirements for hardened state-aware lanes. Both hardened proofs are short-lived and single-use.

Lane	Proof	Binding tuple
W2H	`authzReuseGrant`	ACME account, normalized identifier set, prior-certificate fingerprint, reusable authorization state.
W3H	`pqReenrollmentToken`	ACME account, normalized identifier set, prior-certificate fingerprint, mTLS possession of the prior certificate.

at a separate local inspection site. The `http-01` callback path is exposed back to the CA by reverse SSH forwarding of the challenge service, with an observed RTT of approximately 39.10 ms. This slice tests whether the state-aware advantage persists beyond the single-host runner and the emulated network surface.

Table 3: Harvested studies used in the paper. All planned cells in the studies reported here completed successfully.

Study	Cells	Purpose
Pilot	54	Validate workflow purity and instrumentation.
Wide campaign	96	Main enrollment-plane map for W0/W2/W3.
Hostile edge	3	Stress slice at C5/N4/L2.
Migration	12	Scenario-cost study for S4.
Hardening	90	Security-performance slice for W2H/W3H.
WAN capstone	6	Selective real-path validation.

5 MEASUREMENT RESULTS

5.1 The Wide Enrollment Plane

Figure 2 shows the main enrollment-plane result. W0 is consistently the slow path. Its median latency rises from 4866.96 ms in N0 to 9477.78 ms in N4. W2 stays far below it, with medians of 119.37 ms, 820.32 ms, 1558.22 ms, and 2839.18 ms across N0, N2, N3, and N4. W3 is even faster in the local regime, with 70.28 ms in N0, then

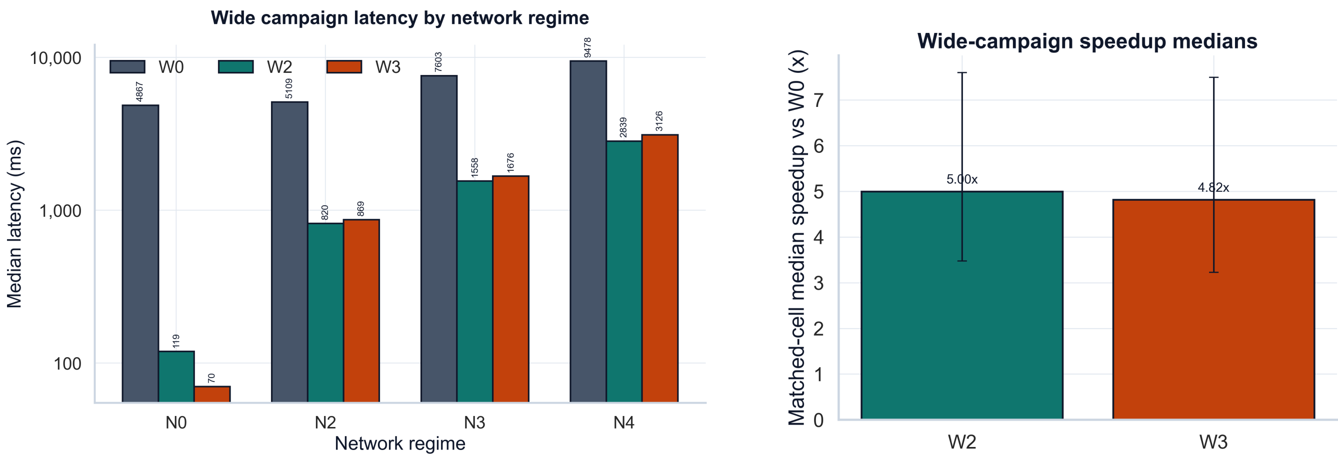

Figure 2: Wide-campaign view of the enrollment plane. Left: median latency by workflow and network regime. Right: matched-cell median speedups versus W0. Both state-aware lanes dominate fresh bootstrap issuance, but their internal ranking changes with regime.

Table 4: Crypto, network, and load regimes. The network regimes are emulated in the controlled campaign; the WAN capstone is separate.

Profile	Meaning
C0	Classical P-256 baseline.
C1	OQS Dilithium2 profile.
C3	OQS Falcon-512 profile.
C5	OQS Dilithium5 profile.
N0	Near-local network: 1 ms RTT, no loss, 1000 Mbps.
N2	Moderate regime: 80 ms RTT, 5 ms jitter, 200 Mbps.
N3	High-latency regime: 160 ms RTT, 10 ms jitter, 100 Mbps.
N4	Hostile edge: 300 ms RTT, 20 ms jitter, 0.5% loss, 50 Mbps.
L0	Isolated enrollment.
L1	Bursty operation with concurrency 8.
L2	Hostile-edge burst with concurrency 32.

reaches 868.85 ms, 1676.01 ms, and 3125.53 ms as the network becomes harsher.

The first-order result answers RQ1: state-aware lanes substantially reduce enrollment cost relative to fresh bootstrap issuance across the main measured surface.

Table 5: Matched-cell speedup summary versus W0 in the wide campaign. The medians show that both state-aware lanes dominate bootstrap issuance, with no global ranking independent of regime.

Workflow	Matched cells	Median	p95	Min	Max
W2	32	5.00x	47.37x	1.01x	70.26x
W3	32	4.82x	96.02x	1.72x	133.27x

Table 5 gives the matched-cell speedup summary. W2 achieves a median speedup of 5.00x versus W0, while W3 reaches 4.82x. The

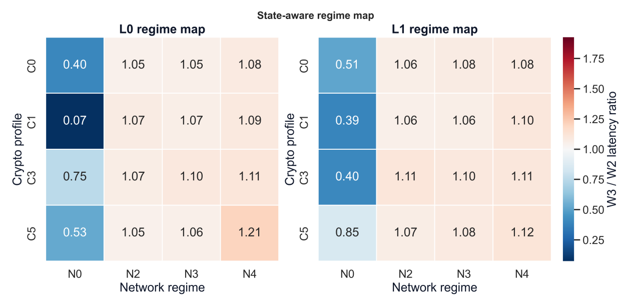

Figure 3: W2/W3 regime map. Each cell summarizes the median latency ratio between W3 and W2 under matched crypto, network, and load conditions. Values above one indicate regimes where W2 is faster, values below one indicate regimes where W3 is faster.

p95 and maximum values show that W3 has the more dramatic best cases: its strongest matched cell reaches 133.27x, and the fastest absolute campaign cell is W3/C0/N0/L0 at 36.24 ms. Yet W2 has the stronger median across the full surface.

These results answer RQ2 at the campaign level: state-aware enrollment has no single stable winner across the measured surface. The appropriate reading is instead a regime-dependent phase diagram. W3 dominates the most benign cells, whereas W2 emerges as the more robust median choice once the full range of operating conditions is considered.

5.2 Why Request Count Is Not Enough

Across workflow-network medians, W0 carries roughly 99 kB to 101 kB of HTTP payload and about fifty requests with repeated authorization polling. W2 compresses that to about 65 kB and roughly thirty-two requests with zero polls. W3 is leaner still, around 57.9 kB and twenty-seven requests.

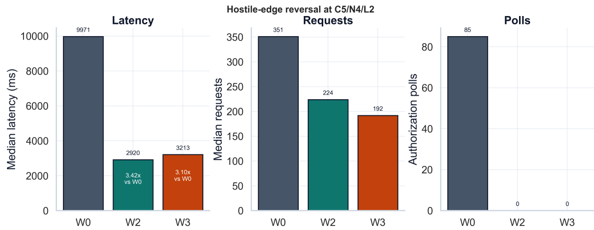

Figure 4: Hostile-edge slice at C5/N4/L2. W3 keeps the leanest control plane, while W2 wins on end-to-end latency.

Taken at face value, the control-plane trace suggests a simple expectation: when both lanes are available, W3 should dominate W2 because it sends fewer requests. The measurements show where that expectation breaks down. In Equation 1, request count contributes the workflow-geometry term, while payload shape, synchronization points, endpoint behavior, and load-induced queueing enter through the remaining terms. The hostile-edge slice makes this separation visible by amplifying the non-request costs far enough to reverse the benign-case ranking.

5.3 Hostile-Edge Reversal

Figure 4 shows the hostile-edge reversal. At C5/N4/L2, W0 takes 9971.26 ms. W2 takes 2919.51 ms, a 3.42x speedup. W3 takes 3212.60 ms, a 3.10x speedup. Both state-aware lanes remain much faster than bootstrap issuance, but W2 overtakes W3.

The reversal is important because W3 retains the leanest control plane: 192 median requests and no authorization polls, compared with 224 requests for W2 and 351 requests plus 85 polls for W0. The hostile-edge slice therefore gives the sharpest empirical answer to RQ2. Under hostile-edge stress, control-plane compactness remains beneficial but loses predictive authority: latency is shaped by the full enrollment path, including payloads, synchronization, endpoint behavior, and contention.

For deployment, lane selection should be policy and regime-aware. A CA or client that supports multiple state-aware paths should choose among them according to available prior state and current operating conditions, with request count treated as one signal within the broader selection policy.

5.4 Migration as Scenario Cost

The migration study evaluates S4 as a composed operational scenario: a client first obtains the classical certificate state needed for later certificate-based fast re-enrollment, then uses that state to enter the fast path. This makes S4 relevant for deployment planning, since real fleets may have to pay the bootstrap cost before they can benefit from W3. Its measurements describe the cost of acquiring and consuming prior state across a migration path, rather than the marginal behavior of a single enrollment lane.

Across the 12 migration cells, S4 has a stable control-plane shape of approximately 17 requests and 2 polls. Latency clusters primarily by network regime. The N2 cells fall between 845.95 ms and 865.26 ms, the N3 cells between 1669.80 ms and 1678.03 ms, and the N4 cells between 3108.93 ms and 3144.44 ms. This clustering occurs even though HTTP bytes vary from roughly 23.1 kB to 83.4 kB. This result supports the semantic separation in Table 1. S4 should be read as a composed migration path for planning and budgeting, while W2 and W3 remain marginal enrollment lanes whose rankings are meaningful within a fixed operating regime.

Table 6: Selective WAN capstone over a Madrid-to-Frankfurt path. Both state-aware lanes preserve a strong advantage over W0 outside the single-host runner.

Workflow	Crypto	Reps	Median time (ms)	Requests	Polls	Speedup vs W0
W0	C0	3	5102.55	11	2	1.00x
W0	C1	3	6619.02	11	2	1.00x
W2	C0	3	805.34	7	0	6.34x
W2	C1	3	953.17	7	0	6.94x
W3	C0	3	904.27	6	0	5.64x
W3	C1	3	897.12	6	0	7.38x

5.5 WAN Capstone

Table 6 reports the selective WAN capstone. The result preserves the same qualitative picture outside the local runner. Under C0, W0 takes 5102.55 ms, while W2 and W3 take 805.34 ms and 904.27 ms. Under C1, W0 takes 6619.02 ms, while W2 and W3 take 953.17 ms and 897.12 ms. The corresponding speedups range from 5.64x to 7.38x.

The WAN slice is scoped as a confirmatory probe. It places the CA and client on a real network path to test whether the state-aware advantage observed in the controlled campaign persists beyond the single-host runner and the emulated network surface. In this setting, the advantage persists, while the relative ordering of W2 and W3 remains sensitive to the measured cell conditions.

6 HARDENING THE STATE-AWARE LANES

6.1 Why Fast Paths Need Fresh Binding

The measurement results establish the operational value of prior state, the hardening problem is to define when that state may be safely consumed. In a naïve state-aware lane, the CA can accept reusable authorization or prior-certificate possession without a fresh proof bound to the current transaction. Such acceptance grants excessive authority to stale or weakly bound state, allowing old material to be replayed, presented under the wrong account, applied to a different identifier set, or detached from the prior certificate that justified the fast path.

The hardened design makes this state dependency explicit. A client may benefit from prior state by presenting a fresh, single-use proof that binds the consumed state to the current ACME account, normalized identifier set, and prior certificate.

Table 7 summarizes the design and performance cost. W2H uses `authzReuseGrant` to authorize consumption of reusable authorization state. W3H uses `pqReenrollmentToken` to authorize `/pq-order` re-enrollment. Both proofs are short-lived, single-use, and bound to the ACME account, normalized identifier set, and prior-certificate fingerprint. The client requires the corresponding hardened directory advertisement, preventing silent fallback to a weaker lane.

Table 7: Naïve versus hardened state-aware lanes. Speedups are computed against the slice-median W0 baseline, overheads are matched-cell hardened-over-naïve ratios.

Lane	Fresh proof	Binding	Security effect	Median	Speedup	Overhead
W2	none	reusable authz only	Reuses authorization state without fresh transaction proof.	1555.44 ms	5.03x	–
W2H	`authzReuseGrant`	acct + ids + cert fp	Rejects replay, expiry, account mismatch, identifier mismatch, and prior-state mismatch.	1713.48 ms	4.57x	1.102x
W3	none	prior cert over mTLS	Uses prior-certificate possession without a fresh transaction token.	1506.35 ms	5.19x	–
W3H	`pqReenrollmentToken`	acct + ids + cert fp	Rejects replay, expiry, account mismatch, identifier mismatch, and prior-certificate mismatch.	1667.42 ms	4.69x	1.105x

Table 8: Representative negative scenarios for the hardened state-aware lanes.

Scenario	W2H	W3H	Evidence
Replay after first use	reject	reject	Pebble unit tests
Expired proof	reject	reject	Pebble unit tests
Account mismatch	reject	reject	Pebble unit tests
Identifier mismatch	reject	reject	Pebble unit tests
Prior-cert mismatch	reject	reject	Pebble unit tests
Missing active reusable state	reject	n/a	Pebble unit tests + executable model

6.2 Executable Security Evidence

Table 8 gives the representative negative scenarios exercised by the artifact. The hardened variants reject proof replay, expired proofs, account mismatch, identifier mismatch, prior-certificate mismatch, and inactive prior state. The naïve variants remain as explicit baselines because the security-performance question depends on comparing permissive and hardened admission surfaces.

The executable model enumerates 9,216 scenarios spanning W0, W2, W2H, W3, W3H, and S4, with 2,164 accepted overall. The security signal lies in the accepted-state distribution. Naïve W2 admits 192 states, compared with 12 for W2H; naïve W3 admits 32 states, compared with 8 for W3H. Hardening therefore removes 180 W2 states and 24 W3 states that remain admissible in the naïve variants. These removed states correspond to the stale, replayed, or weakly bound cases that the fresh state-bound proofs are designed to close.

Symbolic validation. The artifact also includes a reduced Tamarin model scoped to the freshness-and-binding core of the hardened lanes. The model abstracts the state transitions governing W2H and W3H. The archived proof run verifies eight lemmas: authenticity and non-replay for both hardened lanes, existential witness traces for successful hardened executions, and witness traces showing that the naïve baselines remain executable without fresh proof objects. As a symbolic complement to the executable model, this layer independently checks the freshness and binding obligations introduced by hardened state-aware enrollment, while leaving full end-to-end ACME verification outside its scope.

6.3 Performance Cost of Hardening

Figure 5 shows the security-performance frontier. In the selective hardening slice, W0 has a median of 7824.85 ms. The naïve state-aware lanes are much faster: 1555.44 ms for W2 and 1506.35 ms for W3. Hardening raises those to 1713.48 ms and 1667.42 ms. The matched-cell median overhead is 1.102x for W2H over W2 and 1.105x for W3H over W3.

This answers RQ3. Fresh, state-bound proofs are not free, but they preserve most of the operational value of state-aware acceleration. The hardened lanes retain 4.57x and 4.69x speedups over the slice-median W0 baseline while sharply narrowing the accepted state surface.

7 DISCUSSION

State changes the measurement problem. The central lesson is that post-quantum ACME enrollment is structured by protocol state as much as by cryptographic profile. A client renewing through reusable authorization and a client re-enrolling with a prior mTLS certificate occupy different regions of the enrollment state space. Each lane exposes a distinct timing structure to the network and carries its own security assumptions.

State-aware rankings are regime-dependent. W3 applies the strongest control-plane compression in the measured design space and achieves the fastest benign cells. Across the wide campaign, however, W2 delivers the stronger median and wins the hostile-edge slice. The ranking changes with the operating regime. Deployment policy should select among lanes according to available prior state, current conditions, and security posture, rather than hard-code a single fast path.

Migration paths and marginal lanes answer different questions. S4 captures the cost of reaching a state in which certificate-based fast re-enrollment can be used. This makes it a deployment-planning scenario rather than a marginal enrollment lane. Operators therefore need both measurements: the marginal cost once prior state exists, and the scenario cost of acquiring and consuming that state during migration.

Fast paths require explicit authority. The security result is that state-aware acceleration expands the admission surface. Once a CA accepts prior state as authority, freshness and binding become part of the protocol contract. The hardened lanes provide a practical design point: require a fresh proof bound to the ACME account, the normalized identifier set, and the prior certificate, while preserving most of the speedup that motivated the fast path.

Policy implication. The measurements support a regime-aware enrollment policy. W0 remains the bootstrap lane when no reusable state exists. W2 or W2H provides a robust renewal path when reusable authorization is available, especially under harsher regimes. W3 or W3H can deliver the strongest absolute performance when a prior certificate exists and conditions are benign. For any state-aware lane exposed as a supported CA interface, the hardened variant is the safer default.

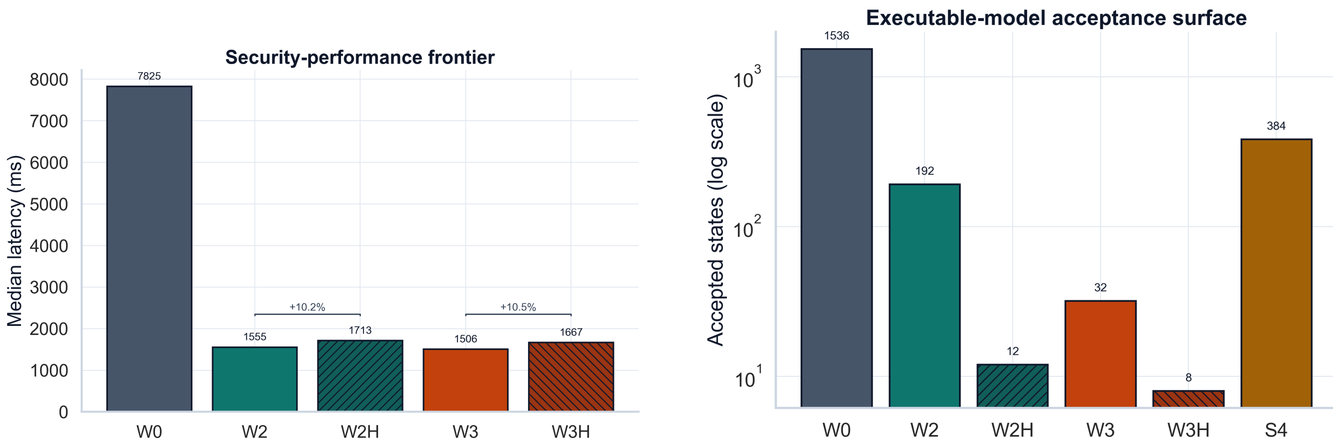

Figure 5: Security-performance frontier of state-aware hardening. Left: median latencies in the selective security slice. Right: accepted-state counts in the executable model, on a log scale. Hardening sharply narrows the admissible state surface while retaining most of the performance advantage over W0.

8 THREATS TO VALIDITY

Controlled network surface. Most measurements use a controlled runner with emulated network and load regimes. This surface enables matched-cell comparison across workflows, profiles, and regimes, and should therefore be read as a laboratory evaluation rather than a global Internet measurement campaign. The WAN capstone adds a real-path check that limits a purely local interpretation of the results, while remaining selective in scope.

Implementation basis. The artifact is built on modified Pebble and LEGO components, following the implementation lineage of prior post-quantum ACME work [5]. Pebble's role as a test CA bounds the interpretation of the measurements: the results support protocol and systems design claims, not production SLOs for any particular public CA.

Profile coverage. The crypto profiles include a classical baseline and several post-quantum-oriented profiles from the OQS-backed implementation. This set is broad enough to separate workflow and regime effects from a single primitive choice, while leaving many certificate-chain, hybridization, and production-policy variants outside the evaluated space.

Attribution limits. The request logs, phase timings, and system metrics support the paper's claims about workflow geometry and regime shifts. They provide system-level evidence for the observed effects, while stopping short of complete microarchitectural attribution for every millisecond. Accordingly, Equation 1 should be read as an interpretive decomposition of enrollment cost. The hostile-edge reversal is an empirical result, its exact causal decomposition remains future work.

Formal scope. The executable model and reduced symbolic model focus on the freshness-and-binding core of the hardened lanes. Their purpose is to check the replay and binding-mismatch scenarios closed by hardened state-aware proofs. Full formal verification of ACME lies outside this scope.

Artifact packaging. The research artifact contains the harness, modified components, canonical result tables, derived figures, executable model, and reduced symbolic model. Heavy traces and operator logs are packaged separately from the anonymous review bundle.

9 CONCLUSION

This paper studied post-quantum ACME enrollment as a state-aware control plane. The central finding is that enrollment cost is governed by prior state, workflow geometry, network regime, and load, not by cryptographic size alone. Across the wide campaign, reusable-authorization renewal and certificate-based fast re-enrollment both substantially outperform fresh bootstrap issuance. Their ranking, however, changes with regime. The hostile edge gives the sharpest lesson: the leaner control plane does not automatically yield lower end-to-end latency.

The security result follows from the same state-based account of enrollment. State-aware acceleration derives its performance benefit from prior state, and safe consumption of that state requires explicit freshness and binding conditions. The hardened lanes introduced here enforce those conditions, narrowing the admissible state surface while preserving most of the performance advantage over fresh issuance.

The practical conclusion is that post-quantum WebPKI migration must treat ACME enrollment as a primary systems and security concern. Lane selection should follow the available prior state and the operating regime, composed migration paths should be budgeted as scenario costs, and any fast path that consumes prior state should enforce explicit freshness and binding conditions.

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A OPEN SCIENCE

The anonymous artifact supporting this submission is made available at the following URL:

https://anonymous.4open.science/r/acme-friction/README.md

The artifact contains the benchmark harness, the modified Pebble/LEGO sources, the CSV results, the scripts used to regenerate tables and figures, and the executable and reduced symbolic models needed to inspect the paper's core claims.

B ETHICAL CONSIDERATIONS

This work does not involve human subjects, user data, or experiments against third-party production services. The measurements were conducted on controlled infrastructure and on test deployments built from Pebble/LEGO-based components. The WAN capstone traverses an operator-controlled path and exposes only the ACME challenge service required by the experiment.